

Single RNA cotranscriptional folding trajectories from stochastic simulations

Overview

The program `ff-co_trajectory` simulates co-transcriptional RNA folding and produces trajectories that show how the structure changes over time. The program extends the sequence nucleotide by nucleotide and performs a stochastic kinetic folding simulation for each transcript length. Each trajectory is represented as a sequence of structures with associated energies and transition times.

Using FASTA input

You can start a trajectory simulation from a predefined **FASTA file**, which contains a sequence and optionally an initial structure. By giving an initial structure the start length for the cotranscriptional simulation can be specified. When no initial structure is given, the simulations start from transcript length 1.

For example, the file `dld3.fa` contains a designed sequence, and as no structure is given the simulation would start at transcript length 1:

```
>dld3
UCAGUCUUCGCGCUGCGUGUAUCGAUUCGGUUUCAGUUUUUAUUGC
```

Here is an example with a starting length of three from an unfolded structure:

```
>dld3
UCAGUCUUCGCGCUGCGUGUAUCGAUUCGGUUUCAGUUUUUAUUGC
...
```

Input Parameters

You can specify the transcription rate via the extension time (`--t-ext`), which defines the simulation time per transcript length. The posttranscriptional folding time (`--t-end`) defines the simulation time at full sequence length. To express simulation time in seconds, set the rate constant `k0` to `1e6`. To further familiarize yourself with the parameters:

```
ff-co_trajectory --help
```

You can run the trajectory simulation as follows:

```
cat dld3.fa | ff-co_trajectory --k0 1e6 --t-ext 0.02 --t-end 1.0
```

All visited structures are printed with their energies and transition times.

Pausing sites

When you do not want an uniform extension time, but define pausing sites, you can define the positions and the duration of the pauses in two separate input parameters: **p-pos** are the positions of the pausing sites and **t-pos** the respective duration. To have a pause at position 2 for 1s and one at position 10 for 0.5s:

```
cat dld3.fa | ff-co_trajectory --k0 1e6 --t-ext 0.02 --t-end 1.0 --p-pos
2,10 --t-pos 1,0.5
```

Example output

An example simulation output is shown below for a full sequence length:

GCGUUUCCAGGGUUUAGACGGACGGGUGUGACUCGCCCAGCCCCGACCUC	energy	arrival-time
waiting-time mean-waiting		
.....	0.00	0.00000000e0
5.48838308e-1 1.15990269e0		
.....(.....)	2.10	5.48838308e-1
7.42186429e-1 6.00461698e-1		
.....(.....)(.....)	4.50	1.29102474e0
6.62519714e-1 3.57034800e-1		
.....(.....)	2.40	1.95354445e0
8.70308936e-3 5.15932279e-1		
.....	0.00	1.96224754e0
2.24827455e0 1.15990269e0		
.....(.....)	1.90	4.21052209e0
1.08942365e0 4.14255646e-1		
.....((.....))	0.10	5.29994574e0
3.93454939e-1 5.63963354e-1		
.....((.....))	-0.20	5.69340068e0
1.06769149e-1 4.52730587e-1		
.....((.....))	1.80	5.80016983e0
4.33243095e-1 3.16296938e-1		
.....(((.....)))	-2.40	6.23341293e0
1.28813977e0 8.21431625e-1		
.....(((.....)))	-1.60	7.52155270e0
2.45415452e-1 2.40731917e-1		
.....(((.....)))	-1.50	7.76696815e0
6.94202383e-2 4.10851300e-1		
.....((((.....))))	-5.60	7.83638839e0
3.47571920e0 3.26611240e0		
.....((((.....))))	-2.90	1.13121076e1
1.55874453e0 4.13337266e-1		
.....((((.....))))	-5.60	1.28708521e1
2.36328447e0 3.26611240e0		
.....((((.....))))(.....)	-2.10	1.52341366e1

7.41356896e-3	4.84982503e-1		
.....(((.(.((.(.....)).))))((...)).....		-4.70	1.52415502e1
2.00829892e1	1.23961105e1		
..(.....(((.(.((.(.....)).))))((...)).....)		-0.50	3.53245394e1
1.26546682e0	7.99897165e-1		
.....(((.(.((.(.....)).))))((...)).....		-4.70	3.65900062e1
4.58067121e1	1.23961105e1		

Each line represents one **structure** visited during the trajectory, with columns:

Header	Description
sequence	Corresponding structure in dot-bracket notation.
energy	Free energy evaluation (kcal/mol).
arrival-time	Simulation time at which the structure was observed.
waiting-time	Time spent in the structure until transition.
mean-waiting	Mean waiting time in the structure (1/flux)

See also

For cotranscriptional ensemble-level analysis across many trajectories, see the example in [ff-co_timecourse](#), which aggregates population data over multiple stochastic runs.